

# Advanced Production Management (F000896A)

## Project Report 2026

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# Introduction to the case

This project focuses on optimizing production planning in a multi-stage manufacturing system using a Mixed Integer Programming (MIP) approach. The objective is to minimize total costs while ensuring efficient production and inventory decisions. By comparing solutions based on forecasted and realized demand, the project provides insights into the impact of uncertainty and the trade-offs between cost efficiency and service levels.

## Assignment 1: Formulating an MIP model with infinite production capacity

### 1.1 Plan based on forecasted demand

#### 1.1.1 Indices

- $i \in parts = \{B1401, B2302, B3201, B4702, V1501, V6302, E2801\}$
- $t \in weeks = \{1, 2, \dots, 30\}$

#### 1.1.2 Parameters

The model covers  $T = 30$  periods (weeks). Each part  $i$  has a lead time  $LT_i$  (periods between placing and receiving an order), a minimum lot size  $L_i$  (minimum quantity per order) and a beginning inventory  $I_{i,0}$  (initial stock at the start of the horizon). Holding costs  $h_i$  apply per unit per week, while setup costs  $S_i$  are incurred per order placement. Finally,  $FD_{i,t}$  and  $RD_{i,t}$  represent respectively the forecasted and realized demand for part  $i$  in period  $t$ .

While the forecasted and realized demand for the end product ( $E2801$ ) are externally driven, component demand ( $V1501$ ,  $V6302$ ,  $B1401$ , ...) is derived internally from the production schedule of their parent item. Component gross requirements are determined by the Planned Order Releases of the parent, multiplied by the Bill of Materials (BOM) factor. This demand occurs one lead time period before the parent is completed, creating a cascading effect throughout the BOM: as soon as a parent part enters production, the required components are immediately withdrawn from inventory.

| Symbol     | Description                                 |
|------------|---------------------------------------------|
| $T$        | forecast horizon (30 weeks)                 |
| $LT_i$     | lead time of part $i$                       |
| $L_i$      | minimum lot size of part $i$                |
| $I_{i,0}$  | initial inventory of part $i$               |
| $h_i$      | holding cost (€ / unit / week)              |
| $S_i$      | setup cost (€ / setup)                      |
| $FD_{i,t}$ | forecasted demand of part $i$ in period $t$ |
| $RD_{i,t}$ | realized demand of part $i$ in period $t$   |

Table 1: Overview of model parameters

#### 1.1.3 Decision variables

Three decision variables are defined for each part  $i$  and period  $t$ . The first,  $x_{i,t}$ , represents the units ordered or produced, and is continuous and non-negative. The second is  $y_{i,t}$ , is binary, equaling 1 when an order is initiated for part  $i$  in period  $t$ , thereby incurring setup cost  $S_i$ . The third,  $I_{i,t}$ , is a non-negative integer representing the inventory level of part  $i$  at the end of period  $t$ , after all receipts and withdrawals in period  $t$  have been processed. A holding cost  $h_i$  is charged per remaining unit in inventory at the end of each period.

$$x_{i,t} = \text{number of units of part } i \text{ ordered in period } t \quad I_{i,t} = \text{inventory (units) of part } i \text{ in period } t$$

$$y_{i,t} = \begin{cases} 1, & \text{if an order for part } i \text{ in period } t \text{ is placed} \\ 0, & \text{otherwise} \end{cases}$$

### 1.1.4 Model

The objective function minimizes the total cost over 30 weeks across all parts consisting of two cost components. The first is the holding cost  $h_i \cdot I_{i,t}$ , charged for every unit of part  $i$  remaining in inventory at the end of period  $t$ . The second is the setup cost  $S_i \cdot y_{i,t}$ , incurred every time a production run or order is initiated for part  $i$  in period  $t$ . No production cost is included since total quantity is fixed regardless of timing, it would be identical across all feasible solutions and therefore does not affect the optimal solution. The model faces a classic trade-off: large infrequent lots save on setup costs but increase inventory, while small frequent lots do the opposite. The optimizer finds the balance between these two forces.

$$\min \sum_{t=1}^{T=30} \sum_{i \in \text{parts}} (h_i I_{i,t} + S_i y_{i,t})$$

### 1.1.5 Constraints

The model has four sets of constraints. The inventory balance constraint ensures that end-of-period stock is consistent with what was available before, what arrived, and what was consumed. End-of-period stock for part  $i$  equals the stock from the previous period, plus any order  $x_{i,t-LT_i}$  placed  $LT_i$  periods earlier (so arrives in period  $t$ ), minus the forecasted demand  $FD_{i,t}$ . When  $t - LT_i < 1$ , no order placed within the planning horizon could have arrived yet, so  $x_{i,t-LT_i}$  drops out.

$$I_{i,t} = I_{i,t-1} + x_{i,t-LT_i} - FD_{i,t} \quad \text{if } t - LT_i < 1 \Rightarrow x_{i,t} = 0 \Rightarrow I_{i,t} = I_{i,t-1} - FD_{i,t}$$

The second constraint links ordering quantity to the setup variable: production is only allowed when a setup takes place, enforced by  $x_{i,t} \leq M \cdot y_{i,t}$ . If  $y_{i,t} = 0$ , then  $x_{i,t}$  is forced to zero; if  $y_{i,t} = 1$ , production is allowed up to big  $M$ . The value of the big  $M$  was set to 25 times the total forecasted demand over all 30 periods, slightly above the highest BOM multiplier in the system (21, or  $3 \cdot 7$ , for B4702), ensuring  $M$  is never restrictive for any part. Note that  $M$  could also be made part-specific, using the exact BOM multiplier per part giving tighter bounds and potentially improve solver performance.

$$x_{i,t} \leq M y_{i,t}$$

The third constraint enforces the minimum lot size. When an order is placed, at least  $L_i$  units must be ordered. Together the second and third constraint ensure that  $x_{i,t}$  is either exactly zero or somewhere between  $L_i$  and  $M$ , with a setup cost charged whenever the latter is the case.

$$x_{i,t} \geq L_i y_{i,t}$$

### 1.1.6 Optimal solution and different cost types

The model reaches a total cost of €394,454, split between €79,500 in setup costs (20%) and €314,954 in holding costs (80%). The dominance of holding costs is unsurprising given the large beginning inventories across all parts.

The production schedule confirms the classic lot-sizing trade-off: Gurobi prefers a small number of large batches, producing every component exactly 3 times and E2801 four times over 30 weeks. Paying one large setup cost once is simply cheaper than paying it repeatedly.

For E2801 the inventory gradually decreases between production runs, hitting exactly zero at the end of weeks 23 and 30. Gurobi avoids holding unnecessary stock. The components follow a similar but more extreme pattern. Their beginning inventories are consumed within the first few weeks, after which inventory drops to zero and stays there. New stock arrives just in time for the next production run of the parent part, with no buffer in between. V6302 illustrates this perfectly: its entire beginning inventory of 1,950 units is consumed in week 1 by the first E2801 lot of 650 units, which requires exactly  $3 \times 650 = 1,950$  units. Not a single unit left over.

The €79,500 in setup costs comes from 3 setups for each of the 6 component parts and 4 setups for E2801, each charged at their respective setup cost from the parameter table.

## 1.2 Evaluation of the forecast-based plan under realized demand

In part 1b, the production plan from part 1a is kept fixed and evaluated under realized demand. No re-optimization is performed; only the end-product demand is replaced by realized values. Unmet demand is backordered at a cost of €250 per unit per period.

The total cost increases from €394,454 to €447,792 (+€53,338), mainly driven by backorder costs of €46,750. The holding cost rises slightly from €314,954 to €321,542, while setup costs remain unchanged at €79,500. The service level equals 93.33% (28/30 periods without backorder) and the fill rate is 96.89% (5,830/6,017 units on time).

Backorders occur only in weeks 23 and 30 (77 and 110 units respectively), showing that the plan performs reasonably well under demand uncertainty but remains vulnerable when realized demand exceeds the inventory buffer.

| Cost comparison   | Part 1a (€)     | Part 1b (€)     |
|-------------------|-----------------|-----------------|
| Holding cost      | €314,954        | €321,542        |
| Setup cost        | €79,500         | €79,500         |
| Backorder cost    | –               | €46,750         |
| <b>Total cost</b> | <b>€394,454</b> | <b>€447,792</b> |

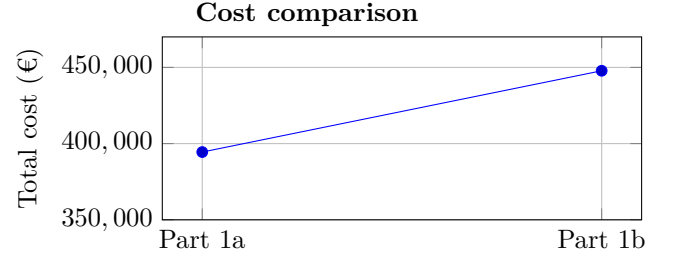


Table 2: Cost comparison between part 1a and part 1b

## Assignment 2: Adjusting the MIP model with finite production capacity

The objective of assignment 2 is to extend the model from assignment 1 by incorporating finite production capacity. In contrast to assignment 1, where production capacity was assumed to be unlimited, the production system is now constrained by the availability of two workstations.

### 2.1 Plan based on forecasted demand

#### 2.1.1 New capacity constraints

Starting from the model in assignment 1a, two additional capacity constraints are introduced, one for each workstation. The inventory balance constraint, setup linking constraint and minimum lot size constraint from assignment 1a remain unchanged. The additional restrictions only limit the maximum feasible production quantities in each period.

Workstation X is used for the production of the end product *E2801*. Its maximum capacity equals 800 units per week. Therefore, the number of units of *E2801* started in production in a given week cannot exceed this weekly capacity:

$$x_{E2801,t} \leq 800 \quad \forall t \in \{1, \dots, 30\} \quad (4)$$

Workstation Y is used for parts *B1401* and *B2302*. Producing one unit of *B1401* requires 3 minutes of processing time, while producing one unit of *B2302* requires 2 minutes. The workstation operates continuously during the week, except for an 80-minute maintenance break. Hence, the available capacity is:

$$7 \cdot 24 \cdot 60 - 80 = 10,000 \text{ minutes per week}$$

The corresponding capacity constraint for workstation Y is:

$$3x_{B1401,t} + 2x_{B2302,t} \leq 10,000 \quad \forall t \in \{1, \dots, 30\} \quad (5)$$

The complete model for assignment 2a is therefore the model from assignment 1a, extended with constraints (4) and (5). These capacity constraints restrict the timing of production and prevent the optimizer from producing arbitrarily large batches in a single period.

### 2.1.2 Cost comparison

The introduction of finite capacity increases total cost from €394,454.00 to €540,488.80. Setup costs rise from €79,500.00 to €145,500.00 (+83%), as the model can no longer concentrate production in a few large batches during the most cost-efficient periods and must spread production over more periods with more frequent, smaller runs. Holding costs increase from €314,954.00 to €394,988.80 (+25.4%), since capacity constraints force earlier production, building up inventory that must be carried over time. Although this results in a higher total cost, it produces a more realistic and operationally feasible production plan.

| Cost comparison   | Part 1a (€)        | Part 2a (€)        |
|-------------------|--------------------|--------------------|
| Holding cost      | €314,954.00        | €394,988.80        |
| Setup cost        | €79,500.00         | €145,500.00        |
| Backorder cost    | –                  | –                  |
| <b>Total cost</b> | <b>€394,454.00</b> | <b>€540,488.80</b> |

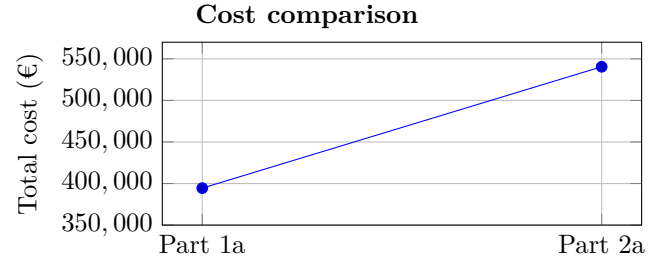


Table 3: Cost comparison between assignment 1a and assignment 2a

## 2.2 Plan used on realized demand

In part 2b, the production plan obtained in part 2a is evaluated using the realized demand instead of the forecasted demand. The production schedule remains unchanged, meaning that no re-optimization is performed. As a result, the setup cost stays identical at €145,500.00. The holding cost increases slightly from €394,988.80 to €401,114.80 due to deviations between forecasted and actual demand. In addition, backorder costs of €27,500.00 arise, indicating that the plan is not able to fully absorb demand variability under capacity constraints.

The service performance of the plan can be measured through the service level and the fill rate. The service level is 96.67%, meaning that in 29 out of the 30 weeks there is no backorder at the end of the period. The fill rate equals 98.17%, which indicates that almost all demanded units are delivered on time. Hence, the plan performs very well in terms of service, despite being constructed under capacity limitations.

Backorders only occur in week 30, where a shortage of 110 units arises. This shows that the plan is generally robust, but still vulnerable in periods where demand peaks exceed the available capacity and inventory buffers.

Overall, the total cost increases from €540,488.80 to €574,114.80. Compared to assignment 1b, the backorder cost is lower, suggesting that the finite capacity plan is slightly more robust against demand fluctuations.

However, shortages still occur, showing that capacity constraints limit the flexibility to respond to higher-than-expected demand.

| Cost comparison   | Part 2a            | Part 2b            |
|-------------------|--------------------|--------------------|
| Holding cost      | €394,988.80        | €401,114.80        |
| Set-up cost       | €145,500.00        | €145,500.00        |
| Backorder cost    | –                  | €27,500.00         |
| <b>Total cost</b> | <b>€540,488.80</b> | <b>€574,114.80</b> |

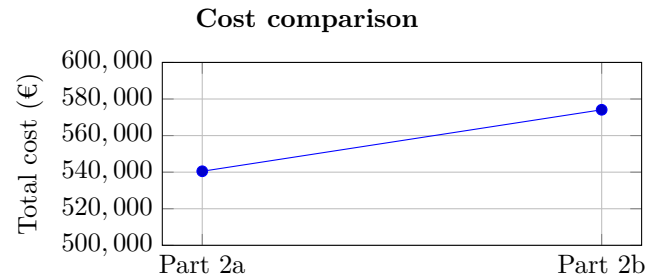


Table 4: Comparison part 2a and 2b

## Assignment 3: Adjusting the MIP model to overtime

### 3.1 Plan based on forecasted demand

In this assignment, the model from Assignment 2a is extended by introducing the possibility of overtime, also called temporary capacity expansion. Unlike the fixed capacity constraints in Assignment 2, this extension allows the company to increase capacity in specific periods at an additional cost, thereby improving flexibility.

### 3.1.1 Decision variables

We add two decision variables to the objective function that will represent the overtime for each workstation.

- $OT_{X,t}$ : additional capacity in units for workstation X in period  $t$
- $OT_{Y,t}$ : additional capacity in hours for workstation Y in period  $t$

### 3.1.2 Parameters

- $c_{OTX}$ : €2 per additional unit of capacity for workstation X
- $c_{OTY}$ : €120 per additional hour of capacity for workstation Y

### 3.1.3 Objective function

$$\min \sum_{t=1}^{T=30} \sum_{i \in \text{parts}} (h_i I_{i,t} + S_i y_{i,t} + c_{OTX} OT_{X,t} + c_{OTY} OT_{Y,t})$$

### 3.1.4 Constraints

The first three constraints remain identical to the base model from Assignment 1a:

- Constraint 1: Inventory balance constraint
- Constraint 2: Setup linking constraint
- Constraint 3: Minimum lot size constraint

The overtime model adds two additional types of constraints.

#### Constraint 4: Capacity constraints with overtime

For workstation X, the production of the end product  $E2801$  in each period cannot exceed the regular weekly capacity of 800 units plus the additional overtime capacity  $OT_{X,t}$ .

$$x_{E2801,t} \leq 800 + OT_{X,t} \quad \forall t \in \{1, \dots, 30\}$$

For workstation Y, the total processing time of parts  $B1401$  and  $B2302$  cannot exceed the regular weekly capacity of 10,000 minutes plus the overtime capacity. Since  $OT_{Y,t}$  is expressed in hours, it is multiplied by 60 to convert it into minutes.

$$3x_{B1401,t} + 2x_{B2302,t} \leq 10,000 + 60OT_{Y,t} \quad \forall t \in \{1, \dots, 30\}$$

#### Constraint 5: Maximum overtime limits

The overtime variables are also bounded. Workstation X can use at most 300 additional units of capacity per week, while workstation Y can use at most 38 additional hours per week.

$$OT_{X,t} \leq 300 \quad OT_{Y,t} \leq 38 \quad \forall t \in \{1, \dots, 30\}$$

### 3.1.5 Cost comparison

The introduction of overtime reduces the total cost by approximately €19,000 (-3.6%) compared to production without overtime. This reduction is driven by the increased flexibility in capacity, which allows the system to better respond to fluctuations in demand. Additionally, the need for inventory buffers is reduced, as production can be more closely aligned with demand peaks instead of relying on early production and stock buildup. Additionally, note that the Gurobi optimizer also identified an alternative optimal solution with the same total cost. This solution differs slightly in the distribution between holding costs and overtime costs, reflecting a different production schedule. However, since the total cost remains identical, we retain this solution.

| Cost Comparison   | Part 2a            | Part 3a            |
|-------------------|--------------------|--------------------|
| Holding Cost      | €394,988.80        | €369,328.80        |
| Set-up Cost       | €145,500.00        | €145,500.00        |
| Backorder Cost    | –                  | –                  |
| Overtime Cost     | –                  | €6,442.00          |
| <b>Total Cost</b> | <b>€540,488.80</b> | <b>€521,270.80</b> |

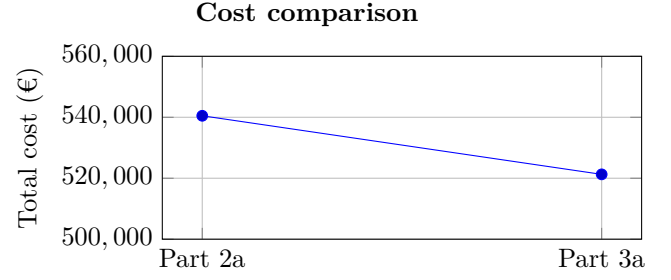


Table 5: Comparison part 2a and 3a

### 3.2 Plan used on realized demand

In part 3b, the production plan obtained in part 3a is evaluated using the realized demand instead of the forecasted demand. The production schedule and the overtime decisions remain unchanged, meaning that no re-optimization is performed. As a result, the setup cost stays identical at €145,500.00 and the total overtime cost remains €6,442.00. The holding cost increases from €369,328.80 to €375,610.80 due to deviations between forecasted and realized demand. In addition, backorder costs of €34,000.00 arise, indicating that the plan is not able to fully absorb demand variability, even with the overtime capacity included in the original plan.

The service performance of the plan can be measured through the service level and the fill rate. The service level is 93.33%, meaning that in 28 out of the 30 weeks there is no backorder at the end of the period. The fill rate equals 97.74%, which indicates that 5,881 out of 6,017 demanded units are delivered on time. Hence, the plan still performs well in terms of service, but realized demand causes shortages in some periods.

Backorders occur in week 26 and week 30, with shortages of 26 and 110 units respectively. This shows that the overtime plan improves production flexibility compared to the strict finite-capacity case, but it does not completely eliminate the effect of demand uncertainty.

Overall, the total cost increases from €521,270.80 to €561,552.80, corresponding to an increase of €40,282.00. This increase is mainly caused by the introduction of backorder costs and a higher holding cost, while setup and overtime costs remain unchanged.

| Cost comparison   | Part 3a            | Part 3b            |
|-------------------|--------------------|--------------------|
| Holding cost      | €369,328.80        | €375,610.80        |
| Set-up cost       | €145,500.00        | €145,500.00        |
| Overtime cost     | €6,442.00          | €6,442.00          |
| Backorder cost    | –                  | €34,000.00         |
| <b>Total cost</b> | <b>€521,270.80</b> | <b>€561,552.80</b> |

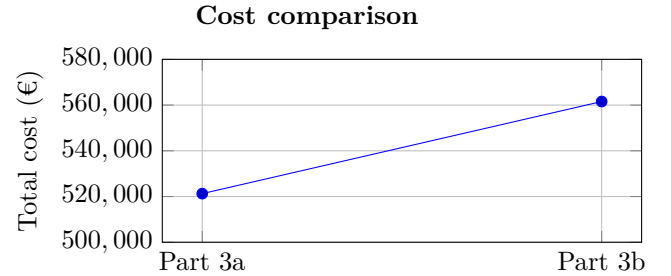


Table 6: Cost comparison part 3a and 3b

#### 3.2.1 Additional sensitivity insight

The previous evaluation assumes that the production plan and the overtime decisions from part 3a remain fixed when realized demand occurs. Under this assumption, backorders still arise in weeks 26 and 30, leading to a backorder cost of €34,000.00. However, an additional sensitivity analysis was performed to examine what would happen if overtime capacity could be adjusted after observing realized demand.

This scenario requires an important operational assumption. It assumes that the realized demand is known sufficiently early, for example at the beginning of the week, and that the company is flexible enough to still adapt overtime decisions before production is executed. This is not the same as assuming instantaneous reaction with zero time delay, which would be unrealistic. Rather, it represents a short-term flexible planning environment in which overtime can still be scheduled in response to updated demand information.

Under this assumption, the remaining overtime capacity is sufficient to absorb the demand deviations. The model can eliminate all backorders by using additional overtime, resulting in a service level and fill rate of 100%. In this sensitivity scenario, the total cost equals €538,767.00. This is higher than the forecast-based 3a cost of €521,270.80, but lower than the fixed-plan realized-demand evaluation in 3b, which resulted in a total cost of €561,552.80. The

improvement is mainly caused by eliminating the €34,000.00 backorder cost, while only €442.00 of overtime cost is required.

This indicates that overtime capacity has value not only as a planned capacity expansion, but also as a short-term flexibility buffer against demand uncertainty. Nevertheless, this interpretation depends strongly on the assumption that overtime decisions can still be adjusted after demand becomes known. If overtime must be fixed in advance together with the production plan, then the original 3b evaluation remains the more realistic comparison.

For a more detailed insight into the results, we refer to the Excel file `OUTPUT.xlsx`.

## Assignment 4: Adjusting the MIP model with machine expansion

In this chapter, the production planning model is extended to allow the company to permanently increase the capacity of its two workstations through machine modernisation. Unlike the overtime option explored in Assignment 3, this expansion represents a one-time capital investment that raises the capacity ceiling for every period in the planning horizon. The goal is to determine the cost-optimal level of expansion and evaluate whether the investment pays off under both forecasted and realised demand.

### 4.1 Plan based on forecasted demand

For Assignment 4a, we build upon the finite capacity MIP model developed in Assignment 2a, which already incorporates the production planning constraints for both workstations alongside the inventory balance, minimum lot size, and Big-M constraints. To extend this model with the possibility of permanent capacity expansion, two new decision variables and four new parameters are introduced.

#### 4.1.1 Decision variables

- $dx$ : represents the number of additional units permanently added to the capacity of Workstation X, and is bounded between 0 and 200.
- $dy\_pct$ : represents the number of 1%-increment steps by which the capacity of Workstation Y is permanently increased, bounded between 0 and 40, corresponding to a maximum expansion of 40%.

#### 4.1.2 Parameters

- $COST_X^{EXP}$ : €10 cost per additional unit of capacity added to Workstation X.
- $COST_Y^{EXP}$ : €1500 cost per 1% increment of additional capacity on Workstation Y.

#### 4.1.3 Objective function

The two new decision variables are independent of the period, since we determine once at the beginning how much the capacity increase will be. In our objective function, we multiply this amount of minutes with the cost per minute.

$$\min \sum_{t=1}^{T=30} \sum_{i \in parts} (h_i I_{i,t} + S_i y_{i,t}) + COST_X^{EXP} \cdot dx + COST_Y^{EXP} \cdot dy\_pct$$

#### 4.1.4 Constraints

Constraints 1–3 (inventory balance, setup linking, minimum lot size) remain unchanged from Assignment 1a.

##### Constraint 4: Capacity constraints with expansion

Two capacity constraints are introduced, one for each workstation. These constraints ensure that production in each period does not exceed the available capacity, which can now be permanently increased through expansion decisions.

- *Workstation X (end product E2801)*:  
The production of the end product in each period is limited by the base capacity of 800 units, which can be increased by the expansion variable  $dx$ .

$$x_{E2801,t} \leq 800 + dx \quad \forall t \in \{1, \dots, 30\}$$

- *Workstation Y (components B1401 and B2302):*

The total processing time required for component production is constrained by the available capacity in minutes. This capacity can be increased proportionally through the expansion variable  $dy\_pct$ .

$$3x_{B1401,t} + 2x_{B2302,t} \leq 10,000 + 100 \cdot dy\_pct \quad \forall t \in \{1, \dots, 30\}$$

#### Constraint 5: Expansion limits

These constraints ensure that the expansion decisions remain within feasible and predefined bounds. The capacity expansion of workstation X is limited to at most 200 additional units, while workstation Y is limited to at most 40%, expressed in discrete percentage steps.

$$0 \leq dx \leq 200 \quad 0 \leq dy\_pct \leq 40$$

#### 4.1.5 Cost comparison

Part 2a results in the highest total cost of €540,488.80 due to high holding and setup costs under strict capacity constraints. In Part 3a, allowing overtime reduces the total cost to €521,270.80 by lowering holding costs, although it introduces additional overtime costs. Part 4a achieves the lowest total cost of €520,723.30 by enabling permanent capacity expansion. The optimization model identifies an ideal expansion of 102 additional units for workstation X and a capacity increase of 8.3% for workstation Y. Despite the associated investment cost for these permanent upgrades, this configuration further reduces holding and setup costs, making it the most cost-efficient solution overall.

| Cost comparison   | Part 2a            | Part 3a            | Part 4a            |
|-------------------|--------------------|--------------------|--------------------|
| Holding Cost      | €394,988.80        | €369,328.80        | €364,253.30        |
| Set-Up cost       | €145,500.00        | €145,500.00        | €143,000.00        |
| Overtime Cost     | –                  | €6,442.00          | –                  |
| Expansion Cost    | –                  | –                  | €13,470.00         |
| Backorder Cost    | –                  | –                  | –                  |
| <b>Total Cost</b> | <b>€540,488.80</b> | <b>€521,270.80</b> | <b>€520,723.30</b> |

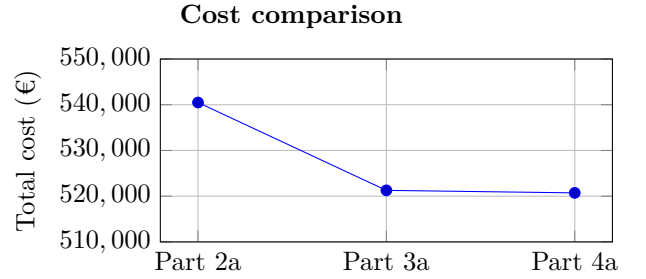


Table 7: Cost comparison part 2a, 3a and 4a

## 4.2 Plan based on realized demand

In part 4b, the production plan obtained in part 4a is evaluated using the realized demand instead of the forecasted demand. Again no re-optimization is performed. As a result, the setup cost stays identical at €143,000.00 and the expansion cost remains €13,470.00. The holding cost increases slightly from €364,253.30 to €370,457.30 due to deviations between forecasted and actual demand. In addition, backorder costs of €30,750.00 arise, indicating that the plan is not able to fully absorb demand variability despite the increased capacity.

The service level is 93.33%, meaning that in 28 out of the 30 weeks there is no backorder at the end of the period. The fill rate equals 97.96%, indicating that the majority of demand is satisfied on time, although slightly less compared to previous scenarios.

Backorders only occur in week 26 and week 30, where shortages of 13 and 110 units arise respectively. This shows that even with expanded capacity, the system remains vulnerable to demand peaks that exceed both capacity and available inventory buffers.

Overall, the total cost increases from €520,723.30 to €557,677.30. This increase is mainly driven by the introduction of backorder costs, while setup and expansion costs remain unchanged. Compared to assignment 2b, the total cost is lower, indicating that capacity expansion helps reduce overall costs. However, shortages still occur, indicating that expansion alone is not sufficient to fully hedge against demand uncertainty.

| Cost comparison   | Part 4a            | Part 4b            |
|-------------------|--------------------|--------------------|
| Holding Cost      | €364,253.30        | €370,457.30        |
| Set-Up cost       | €143,000.00        | €143,000.00        |
| Overtime Cost     | –                  | –                  |
| Expansion Cost    | €13,470.00         | €13,470.00         |
| Backorder Cost    | –                  | €30,750.00         |
| <b>Total Cost</b> | <b>€520,723.30</b> | <b>€557,677.30</b> |

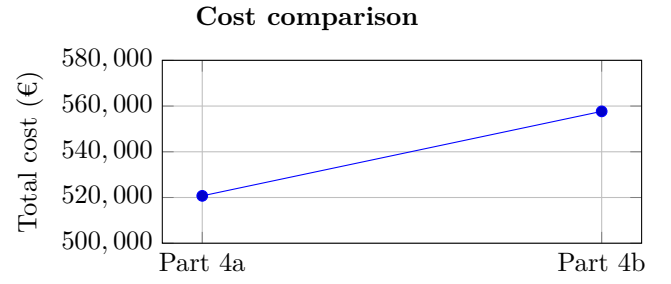


Table 8: Cost comparison part 4a and 4b

## Assignment 5: Adjusting the MIP model with overtime and machine expansion

In this assignment, the MIP model is extended by combining both overtime and permanent capacity expansion. This allows the model to choose between short-term flexibility, overtime, and long-term structural improvements, capacity expansion, or a combination of both.

### 5.1 Plan based on forecasted demand

The model of Part 5 is based on the model of Part 2a. Because Part 5 combines the adjustments of part 3a and 4a, a lot of things return from those models when adding the extra info to the base model.

#### 5.1.1 Parameters

We combine the added parameters from assignment 3 and 4 and we add these to the base model from assignment 2.

- $c_{OTX}$ : €2 per additional unit of capacity for workstation X
- $c_{OTY}$ : €120 per additional hour of capacity for workstation Y
- $COST_X^{EXP}$ : €10 cost per additional unit of capacity added to Workstation X
- $COST_Y^{EXP}$ : €1500 cost per 1% increment of additional capacity on Workstation Y

#### 5.1.2 Objective function

$$\min \sum_{t=1}^{T=30} \sum_{i \in parts} (h_i I_{i,t} + S_i y_{i,t} + c_{OTX} OT_{X,t} + c_{OTY} OT_{Y,t}) + COST_X^{EXP} \cdot dx + COST_Y^{EXP} \cdot dy\_pct$$

#### 5.1.3 Constraints

Constraints 1–3 (inventory balance, setup linking, minimum lot size) and constraints 4–5 (maximum overtime, maximum expansion) remain unchanged from assignments 1a, 3a and 4a respectively.

**Constraint 6:** Maximum capacity adjusted with overtime and permanent expansion for both workstations

This first new constraint ensures that the production of the end product  $E2801$  in each period does not exceed the total available capacity of workstation X, which consists of the base capacity, the temporary increase through overtime  $OT_{X,t}$ , and the permanent capacity expansion  $dx$ .

$$x_{E2801,t} \leq 800 + OT_{X,t} + dx \quad \forall t \in \{1, \dots, 30\}$$

The second constraint limits the total processing time required on workstation Y by considering the processing times of components  $B1401$  and  $B2302$ , while allowing capacity to be increased both temporarily through overtime  $OT_{Y,t}$  and structurally through a percentage-based expansion  $dy\_pct$ .

$$3x_{B1401,t} + 2x_{B2302,t} \leq 10,000 + 60OT_{Y,t} + 100 \cdot dy\_pct \quad \forall t \in \{1, \dots, 30\}$$

### 5.1.4 Cost comparison

We observe that a combination of working overtime and machine expansion decreases the total cost. Next to that, all cost types separately are lower than previous parts. This could possibly be the best production plan so far. However, “total cost” is not our only performance measure. In part 5b, where we also take the real demand into consideration, we will also take a look at the fill rate to compare this production plan to the previous ones.

| Cost comparison   | Part 2a            | Part 3a            | Part 4a            | Part 5a            |
|-------------------|--------------------|--------------------|--------------------|--------------------|
| Holding Cost      | €394,988.80        | €369,328.80        | €364,253.30        | €354,296.60        |
| Set-Up Cost       | €145,500.00        | €145,500.00        | €143,000.00        | €132,000.00        |
| Overtime Cost     | –                  | €6,442.00          | –                  | €600.00            |
| Expansion Cost    | –                  | –                  | €13,470.00         | €23,180.00         |
| Backorder Cost    | –                  | –                  | –                  | –                  |
| <b>Total Cost</b> | <b>€540,488.80</b> | <b>€521,270.80</b> | <b>€520,723.30</b> | <b>€510,076.60</b> |

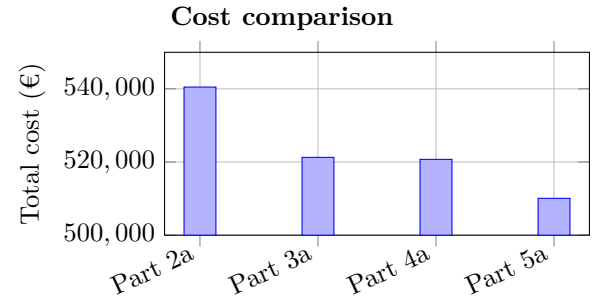


Table 9: Comparison part 2a, 3a, 4a and 5a

## 5.2 Plan based on realized demand

In part 5b, the production plan obtained in part 5a is evaluated using the realized demand instead of the forecasted demand. The production schedule, as well as the chosen capacity expansions and overtime decisions, remain unchanged, meaning that no re-optimization is performed. As a result, the setup cost stays identical at €132,000.00, the expansion cost remains €23,180.00, and the overtime cost stays €600.00. The holding cost increases slightly from €354,296.60 to €360,656.60 due to deviations between forecasted and actual demand. In addition, backorder costs of €37,250.00 arise, indicating that even with both overtime and permanent capacity expansion, the plan cannot fully absorb demand variability.

The service performance of the plan can be measured through the service level and the fill rate. The service level is 93.33%, meaning that in 28 out of the 30 weeks there is no backorder at the end of the period. The fill rate equals 97.52%, which indicates that 5,868 out of 6,017 demanded units are delivered on time. Hence, the plan performs well, but slightly worse compared to some previous scenarios in terms of service robustness.

Backorders only occur in week 26 and week 30, where shortages of 39 and 110 units arise respectively. This shows that even with the combination of overtime and capacity expansion, the system remains sensitive to peak demand periods where available capacity and inventory buffers are insufficient.

Overall, the total cost increases from €510,076.60 to €553,686.60. This increase is mainly driven by the introduction of backorder costs, while setup, expansion, and overtime costs remain unchanged.

Compared with assignment 4b, the total cost is slightly lower, indicating that combining overtime with expansion improves cost efficiency. However, the presence of backorders shows that even this more flexible strategy does not fully eliminate the impact of demand uncertainty.

| Cost comparison   | Part 5a            | Part 5b            |
|-------------------|--------------------|--------------------|
| Holding Cost      | €354,296.60        | €360,656.60        |
| Set-Up Cost       | €132,000.00        | €132,000.00        |
| Overtime Cost     | €600.00            | €600.00            |
| Expansion Cost    | €23,180.00         | €23,180.00         |
| Backorder Cost    | –                  | €37,250.00         |
| <b>Total Cost</b> | <b>€510,076.60</b> | <b>€553,686.60</b> |

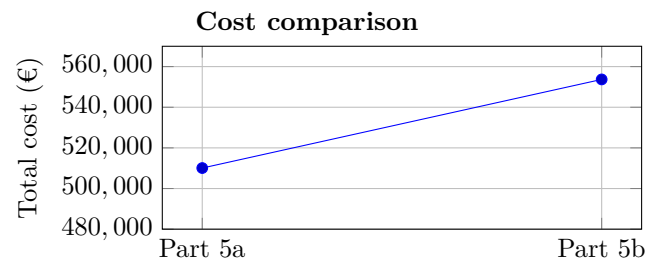


Table 10: Cost comparison part 5a and 5b

## Assignment 6: Adjusting the MIP model with creative ideas

Starting from the model in Part 5, we sought ways to enhance its performance using real demand data, particularly in managing the disparity between forecasted and realized demand, or uncertainty.

## 6.1 6a. Ideas to improve performance and reduce uncertainty

### Idea 1: Working with a safety stock

To account for demand uncertainty, a safety stock was introduced for the end product. In the previous assignments, the production plan was optimized based on forecasted demand only. However, as observed in the dataset, the realized demand consistently deviates from the forecast, which may lead to stockouts and consequently backorder costs. To mitigate this risk, a safety stock policy was implemented.

The safety stock level was determined using a standard statistical approach based on forecast errors. The forecast error per period was calculated as the difference between realized and forecasted demand. The standard deviation of these errors ( $\sigma_e$ ) was then computed to quantify the variability in demand (Limère, 2026):

$$\sigma_e = \text{std}(d_t^{\text{realized}} - d_t^{\text{forecast}}), \quad t \in \{1, \dots, 30\} \quad (1)$$

A first, simplified approach computes the safety stock directly from the forecast error standard deviation:

$$SS = z_{1-\alpha} \cdot \sigma_e \quad (2)$$

This formulation assumes that the safety stock only needs to cover the variability within a single period. However, the system operates with a lead time of  $L = 1$  period for the end product and a weekly review period of  $R = 1$ . The lead time refers to the delay between placing a production order and receiving the finished goods; the review period reflects how frequently production decisions are made. During this combined window of  $L + R = 2$  periods, no corrective action can be taken, meaning the safety stock must absorb all demand fluctuations that occur within this interval. Accounting for this, the standard deviation over the full uncertainty window becomes:

$$\sigma_{LTD} = \sigma_e \cdot \sqrt{L + R} \quad (3)$$

The resulting safety stock is then:

$$SS = z_{1-\alpha} \cdot \sigma_{LTD} \quad (4)$$

This safety stock was incorporated into the MIP model as a lower bound on the inventory of the end product in each period:

$$I_{E2801,t} \geq SS \quad \forall t \in \{1, \dots, T\} \quad (5)$$

By enforcing this constraint, the model is forced to maintain a buffer against demand variability, which shifts production earlier in time and reduces the likelihood of stockouts when realized demand exceeds the forecast. Compared to the model without safety stock, this approach typically leads to higher holding costs due to the additional inventory. However, under realized demand, safety stock effectively buffers against demand variability, significantly reducing backorders and their associated costs. Given the relatively high backorder cost in this problem setting, maintaining a safety stock is economically justified and results in a more robust production plan under uncertainty.

To assess the sensitivity of this approach to the chosen confidence level, the model was solved under two target cycle service levels: 95% ( $z_{1-\alpha} = 1.65$ ,  $SS = 60$  units) and 99% ( $z_{1-\alpha} = 2.33$ ,  $SS = 84$  units). The key results are summarized in Table 11. A full cost breakdown and detailed production schedules can be found in the accompanying Excel file.

Table 11: Safety stock sensitivity analysis: key results

| Design        | Safety Stock | Total Cost (forecast) | Total Cost (realized) | # Backorders | Service level |
|---------------|--------------|-----------------------|-----------------------|--------------|---------------|
| Assignment 5b | (No)         | €510,076.60           | €553,686.60           | 149          | 93.33%        |
| 95% CSL       | 60           | €520,462.60           | €538,728.60           | 50           | 96.67%        |
| 99% CSL       | 84           | €524,617.00           | €536,739.00           | 26           | 96.67%        |

Both safety stock levels significantly reduce backorder costs under realized demand. Notably, the 99% service level yields a lower total cost than the 95% level under realized demand (€536,739 vs. €538,729). This counterintuitive result arises because the backorder penalty (€250/unit/week) vastly exceeds the holding cost (€6/unit/week for E2801). The additional 24 units of safety stock generate approximately €1,850 in extra holding costs but prevent backorders worth

approximately €6,000, yielding a net saving of roughly €4,150. This suggests that for this particular cost structure, even higher service levels could be economically beneficial.

## **Idea 2: Outsourcing production**

### **Motivation and approach**

Part B4702 has both the longest lead time (5 weeks) and the highest BOM multiplier ( $7 \times 3 = 21$  units per end product) of all components. This combination results in large batches that must be ordered well in advance, leading to substantial holding costs. This makes B4702 the most suitable candidate for outsourcing. By sourcing the component from an external supplier capable of faster delivery, the required inventory buffer decreases and holding costs are structurally reduced.

In the proposed scenario, it is assumed that an external supplier can deliver B4702 within 1 week. The internal setup cost of €2,500 per batch is eliminated, but is replaced by an order cost per batch placed with the external supplier. An order cost of €500 per order is used. The robustness of this assumption was verified through a sensitivity analysis.

### **Determining the order cost: sensitivity analysis**

Since no specific supplier data was available, the order cost was not derived using the EOQ formula (see appendix B). That formula is only valid when the EOQ is actually the optimal batch size, a condition that does not hold here given the minimum lot sizes, capacity constraints and BOM structure present in the model [?]. Instead, the impact of the order cost was tested directly by solving the model for values ranging from €250 to €1,500 per order in steps of €250. Table 15 (in appendix B) shows that the total cost difference between €250 and €1,500 per order amounts to only €6,250 over the full planning horizon, which is marginal relative to a total cost of approximately €420,000. Based on this, €500 is used as the central assumption; it is a conservative but realistic estimate that does not materially affect the conclusions.

### **Results compared to Assignment 5**

With a lead time of 1 week and an order cost of €500, outsourcing yields a total cost of €420,963.80 based on forecasted demand, compared to €510,076.60 in Assignment 5a. The reduction of nearly €90,000 is entirely driven by a sharp decrease in holding costs (as intuitively expected given the reduction in lead time). As an additional benefit, the need for permanent capacity expansion on workstation X disappears (Investment X drops from €2,000 to €0), because the increased planning flexibility allows the optimizer to spread production more evenly. (See appendix B: table 16)

### **Sensitivity analysis: effect of lead time**

Since a lead time of 1 week represents an optimistic scenario, the impact of lead times of 2 to 4 weeks was also tested. A lead time of 5 weeks or more is not considered, as this corresponds to the current internal situation and therefore offers no benefit. Table 17 (in appendix B) shows that total cost based on forecasted demand increases as lead time grows longer, but remains well below the level of Assignment 5a in all cases. Even at a lead time of 4 weeks, the saving compared to 5a still amounts to approximately €31,000. Notably, at lead times of 3 or 4 weeks the optimizer reverts to a larger permanent capacity expansion on workstation Y (20% instead of 6.9%) and reinvests in workstation X. This illustrates that a shorter lead time not only reduces holding costs, but also relieves capacity pressure by allowing production to be scheduled more flexibly.

### **Evaluation under realized demand and comparison with 5b**

When the outsourcing plan ( $L = 1$ , order cost = €500) is evaluated under realized demand, total cost increases from €420,864 to €464,574, mainly due to backorder costs of €37,250. The service level (93.3%) and fill rate (97.52%) are identical to those of Assignment 5b (see table 18 in Appendix B). The fact that service level and fill rate are unchanged is logical and worth explaining. The backorders in 5b (and 6b) occurred in weeks 26 and 30, not because B4702 was unavailable in time, but because E2801 inventory was insufficient at those moments to absorb the higher realized demand. In the 5b plan, B4702 stock was already fully depleted from week 6 onwards, the entire supply chain had been drained long before the backorders arose. The bottleneck therefore lay elsewhere in the system. Outsourcing consequently addresses a different problem than demand uncertainty at the level of the end product. This is not a weak conclusion, but rather a sign of critical insight: the measure is cost-efficient, but does not address the root cause of the backorders.

### **Break-even analysis: price premium over internal production cost**

So far it has been assumed that the purchase price at the external supplier equals the internal production cost of B4702. In practice, an external party will typically charge a higher unit price. The break-even analysis therefore calculates up

to what price premium per unit (on top of the internal production cost) outsourcing remains advantageous compared to Assignment 5b. Over the full planning horizon, a total of 78,847 units of B4702 must be purchased. Given a cost saving of approximately €89,000 ( $LT = 1$ ) at equal unit price, the purchase price can be at most €1.13 per unit higher than the internal production cost before the benefit of outsourcing is fully eliminated. For longer lead times the margin is smaller. (See table 19 in appendix B)

### Idea 3: Double exponential smoothing for demand forecasting

One candidate forecasting method to handle demand uncertainty is Holt’s method (double exponential smoothing), which extends simple exponential smoothing by maintaining two smoothing equations: one for the level of the series ( $S_t$ ) and one for the trend ( $G_t$ ), controlled by two smoothing constants  $\alpha$  and  $\beta$  respectively. The  $\tau$ -step-ahead forecast is then given by  $F_{t,t+\tau} = S_t + \tau G_t$ , making it well-suited for data exhibiting a linear trend.

However, applying Holt’s method to this problem is not appropriate for two reasons. First, the method requires historical demand data prior to the planning horizon to reliably initialize  $S_0$  and  $G_0$ , typically through linear regression. This is data that is not available here, as the dataset only covers weeks 1 through 30 with no preceding observations. Second, inspecting the realized demand reveals no clear linear trend; demand fluctuates around a relatively stable mean, making the trend component unnecessary. For these reasons, simpler and more robust approaches are better suited to address the uncertainty between forecasted and realized demand in this setting.

### Idea 4: Increasing the number of periods in the planning horizon

A key observation from the results of Assignments 1 through 5 is that backorders tend to concentrate in the final periods of the planning horizon. This occurs because the optimizer, knowing that no demand exists beyond period 30, deliberately runs down inventory toward the end of the horizon to minimize holding costs. In practice, however, the company will continue to operate beyond 30 weeks, and this so-called *end-of-horizon effect* leads to an artificially optimistic plan that underperforms when evaluated against realized demand.

To mitigate this effect, the planning horizon was extended from 30 to 35 weeks. The additional 5 weeks of demand data serve as a buffer: the optimizer now plans production beyond week 30, preventing it from draining inventory prematurely. All time-dependent costs (setup, holding, backorder, and overtime) over the 35-week horizon are then rescaled to a 30-week equivalent using factor  $D = 30/35$  to allow a fair comparison with the original 30-week results from Assignment 5. Investment costs, being one-time expenditures independent of the horizon length, are not rescaled. A limitation of this approach is that the demand values for weeks 31–35 are necessarily estimated, as no observed data exists beyond the original 30-week horizon. The quality of the results therefore depends on the accuracy of this estimate.

For the estimation, two methods were considered:

- **Linear extrapolation:** A linear least-squares regression was fitted to the 30 observed weeks of demand data. The resulting trend line was then extended to predict demand for weeks 31–35. The realized demand series exhibits a slope of approximately +1.54 units per week, indicating a mild upward trend. This method is statistically grounded but produces extrapolated values where realized demand systematically exceeds the forecast, which can amplify the end-of-horizon backorder effect in the extended weeks.
- **Cyclic repetition:** The demand values from weeks 1–5 are repeated as weeks 31–35. This is a simpler heuristic that assumes demand patterns are roughly periodic. In our dataset, weeks 1–5 represent a low-demand startup phase where realized demand is generally below the forecast, which provides a natural buffer against backorders in the extended period.

A visual comparison of both extrapolation methods can be found in Appendix C.

The key results are summarized in Table 6.1.

On forecasted demand, both extended-horizon variants yield a lower total cost than Assignment 5, confirming that the longer planning horizon allows the optimizer to find more efficient production schedules. The cyclic variant achieves the largest reduction (-3.1%), while the linear variant saves approximately 0.8%.

Under realized demand, however, the two methods diverge significantly. The cyclic variant reduces total cost by 3.3% relative to Assignment 5 (€535,141 vs. €553,687) and cuts backorders from 149 to 67 units, improving the fill rate from 97.52% to 98.89%. The linear variant, by contrast, increases total cost by 9.2% to €604,858 with 91 backorders. This counterintuitive result occurs because the linearly extrapolated realized demand for weeks 31–35 consistently exceeds

| Design            | Total (forecast) | Total (realized) | # Backorders | Service level | Fill rate |
|-------------------|------------------|------------------|--------------|---------------|-----------|
| Assignment 5      | €510,077         | €553,687         | 149          | 93.3%         | 97.52%    |
| Linear extrap.    | €505,814         | €604,858         | 91           | 93.3%         | 98.49%    |
| Cyclic repetition | €494,011         | €535,141         | 67           | 93.3%         | 98.89%    |

Table 12: Extended horizon (35 periods): cost comparison (30-week equivalent)

the extrapolated forecast (by approximately 20–25 units per week), causing the optimizer to under-plan production for those weeks and generating substantial backorders in the extended period. Since all costs, including these artificial backorders, are scaled by factor  $D$ , they inflate the 30-week equivalent cost.

The cyclic variant avoids this issue because weeks 1–5 of our dataset have realized demand below the forecast, so the extended weeks act as a genuine buffer rather than a source of additional backorders.

Based on these results, the cyclic repetition method was selected as the preferred approach for the extended horizon analysis. It demonstrates that by simply extending the planning horizon with a reasonable demand estimate, the system can simultaneously reduce total cost by 3.3%, cut backorders by more than half (from 149 to 67 units), and improve the fill rate from 97.52% to 98.89%. This confirms that a significant portion of the backorder cost observed in Assignment 5 is attributable to the end-of-horizon effect rather than to a fundamental capacity or planning deficiency.

## Idea 5: Capacity expansion both workstations

To investigate whether capacity expansion could mitigate demand uncertainty, sensitivity analyses were conducted for both workstations around the Part 5a optimal capacity levels. A detailed breakdown of all tested values, including full cost decomposition, is provided in Appendix D.

For Workstation X, which directly produces E2801, additional capacity might be expected to reduce backorders. However, backorders remain constant at 149 units for all capacities between 100 and 600 units. Since the Assignment 5a optimizer reached the upper bound of 200 units, we tested higher values, but found no cost improvement. The lowest realized total cost of €553,718 occurs at 200 units, while further expansion only increases investment cost.

| Design  | Max cap X | Forecast cost | Realized cost | Backorders | Service level |
|---------|-----------|---------------|---------------|------------|---------------|
| 1       | 100 u     | €512,123.10   | €557,733.10   | 149        | 93.33%        |
| 2       | 150 u     | €511,953.40   | €555,745.10   | 149        | 93.33%        |
| current | 200 u     | €510,108.10   | €553,718.10   | 149        | 93.33%        |
| 3       | 250 u     | €510,443.10   | €554,053.10   | 149        | 93.33%        |
| 4       | 300 u     | €510,843.10   | €554,453.10   | 149        | 93.33%        |

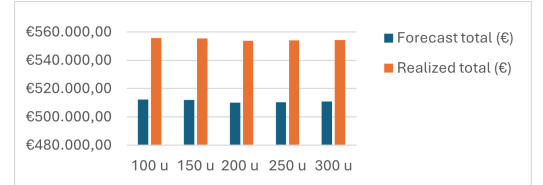


Table 13: Sensitivity analysis: maximum expansion Workstation X

For Workstation Y, forcing a higher expansion produces more B1401 and B2302 components upfront, building a component buffer. However, the backorder cost shows no consistent improvement and even increases at higher expansion levels. The reason is fundamental: Workstation Y only produces intermediate components. No matter how large the component buffer is, if the end product E2801 is not produced at the right time, backorders will still occur.

| Design  | Max cap Y | Forecast cost | Realized cost | Backorders | Service level |
|---------|-----------|---------------|---------------|------------|---------------|
| 1       | 4.1%      | €514,991.30   | €558,601.30   | 149        | 93.33%        |
| 2       | 9.1%      | €512,176.50   | €555,786.50   | 149        | 93.33%        |
| current | 14.1%     | €510,089.40   | €553,699.40   | 149        | 93.33%        |
| 3       | 19.1%     | €511,634.10   | €555,244.10   | 149        | 93.33%        |
| 4       | 24.1%     | €514,752.30   | €558,182.30   | 149        | 93.33%        |

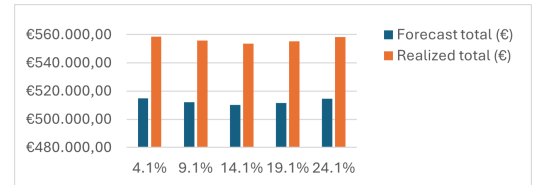


Table 14: Sensitivity analysis: maximum expansion Workstation Y

The root cause is the same in both cases: backorders in weeks 26 and 30 arise from a mismatch in production timing, not a capacity shortage. The optimizer, planning on forecasted demand, does not schedule production close enough to these demand peaks, leaving additional capacity unused if it is not deployed at the right moment.

Both analyses therefore confirm that capacity expansion alone is an ineffective strategy for handling demand uncertainty. The lowest realized cost is achieved at the Part 5a reference values ( $dx = 200$  u,  $dy = 14.1\%$ ). The real solution

lies in planning logic, which motivates the safety stock approach discussed in Idea 1.

## 6.2 Conclusion 6a

Of the five strategies explored, capacity expansion and Holt's forecasting method proved ineffective, the former because backorders stem from a timing mismatch rather than a capacity shortage, the latter due to insufficient historical data. Outsourcing of B4702 achieves large cost savings but does not reduce backorders, since the bottleneck lies at the end-product level. The extended planning horizon reduces both cost and backorders by eliminating the end-of-horizon effect, but requires assumptions about future demand.

The recommended design is **safety stock at the 99% confidence level**. It directly addresses the root cause of uncertainty: forecast errors at the end-product level that propagate into backorders. With a safety stock of 84 units, total realized cost decreases by 3.1% while backorders drop from 149 to 26 units and fill rate improves to 99.57%. Outsourcing and the extended horizon remain valuable as complementary cost-reduction measures, but neither substitutes for an inventory buffer at the point of customer delivery.

## 6.3 6b. Other sources of uncertainty

Although demand uncertainty was the primary focus of this project, a production system faces many additional sources of uncertainty. A useful theoretical lens is the Variability Law: increasing variability always degrades system performance, and since variability propagates through the production chain, reduction efforts should be placed as early as possible. The Buffering Law adds that any system facing variability must be buffered by some combination of inventory, capacity, and time. The models developed in Assignments 1 to 5 rely primarily on inventory as a buffer.

### Process uncertainty

An important internal source of uncertainty is product and process complexity. E2801 has a multi-level bill of materials with seven parts across three production levels, requiring multiple sequential assembly steps. Each step introduces variability, and due to their interdependence, delays or errors propagate through the system. This effect is most critical in Assignments 1 and 2, where no capacity buffer exists to absorb disruptions. Simplifying the BOM or reducing the number of components would shorten lead times and limit cascading variability.

Setup operations further contribute to uncertainty, as each setup affects effective processing time. While reducing setups lowers costs, larger batch sizes can amplify variability propagation, highlighting a trade-off between cost efficiency and operational robustness.

### Capacity uncertainty

Capacity uncertainty arises when available workstation capacity deviates from assumed levels due to machine outages or operational disruptions. In Assignment 2, both workstations operate at fixed capacity with no flexibility, making the system fully exposed to this risk. Assignments 3 and 5 partially mitigate this through overtime flexibility, which acts as a capacity buffer when disruptions occur. Assignments 4 and 5 go further by permanently expanding capacity, lowering baseline utilization and thereby reducing the nonlinear impact of outages.

Introducing rework processes or improving maintenance policies could further reduce the impact of capacity losses by shortening recovery times and stabilizing effective capacity levels.

### Lead time uncertainty

Lead time uncertainty arises when the actual time between ordering and receiving components deviates from assumed fixed values, due to supplier delays, transportation problems, or other disruptions. In this model, all lead times are treated as deterministic across all assignments. However, even modest variation in the lead time of critical components such as B4702 (5 periods) could prevent V6302 production from starting on time, propagating directly to E2801.

None of the models include a safety lead time buffer for purchased components, meaning this risk remains unmitigated across Assignments 1 to 5. Incorporating lead time buffers or working with more reliable suppliers would reduce both the average lead time and its variability, improving system robustness.

### Supply-side uncertainty

Beyond the modeled uncertainties, supply-side disruptions such as supplier reliability issues, quality defects, or yield losses can further impact system performance. Yield losses and rework reduce effective throughput below nominal capacity, tightening bottlenecks and potentially generating shortages that safety stock alone cannot absorb. This highlights the importance of combining different types of buffers (inventory, capacity, and time) rather than relying on a single mechanism.

## References

- Limère, V. (2026). APM3\_ProductionPlanning\_GroupProject [Lecture slides, Advanced Production Management, Ghent University].
- Graves, S. C. (1999). Manufacturing planning and control (Working paper). Massachusetts Institute of Technology.
- Limère, V. (2026). APM4\_Inventory\_Deterministic [Lecture slides, Advanced Production Management, Ghent University].
- Limère, V. (2026). APM5\_Inventory\_Stochastic [Lecture slides, Advanced Production Management, Ghent University].

## Appendix

### Appendix A: EOQ formula

EOQ formula:

$$EOQ = \sqrt{\frac{2DS}{H}}$$

With  $S$  = order cost

*Reference:* Silver, E. A., Pyke, D. F., & Peterson, R. (1998). *Inventory management and production planning and scheduling* (3rd ed.). Wiley.

### Appendix B: Outsourcing tables

#### Outsourcing results

| Order cost (€ / order) | €250        | €500        | €750        | €1000       | €1250       | €1500       |
|------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Total cost (EUR)       | €419,713.80 | €420,963.80 | €422,213.80 | €423,463.80 | €424,713.80 | €425,963.80 |
| # orders B4702         | 5           | 5           | 5           | 5           | 5           | 5           |

Table 15: Outsourcing: order cost sensitivity analysis for B4702 under forecast demand with  $LT = 1$  week

| Cost type         | Part 5a            | Part 6a (outsourcing, LT=1) |
|-------------------|--------------------|-----------------------------|
| Setup cost        | €132,000.00        | €130,500.00                 |
| Holding cost      | €354,296.60        | €273,787.80                 |
| Order cost B4702  | –                  | €2,500.00                   |
| Investment X      | €2,000.00          | €0.00                       |
| Investment Y      | €21,180.00         | €10,380.00                  |
| Overtime X        | €600.00            | €364.00                     |
| Overtime Y        | €0.00              | €3,432.00                   |
| <b>Total cost</b> | <b>€510,076.60</b> | <b>€420,963.80</b>          |

Table 16: Outsourcing: cost comparison between Assignment 5a and 6a under forecast demand

| Lead time (weeks)    | 1 week      | 2 weeks     | 3 weeks     | 4 weeks     |
|----------------------|-------------|-------------|-------------|-------------|
| Total cost 6a        | €420,963.80 | €442,403.80 | €453,828.10 | €478,548.10 |
| vs. LT=1 (best case) | €0.00       | +€21,440.00 | +€32,864.30 | +€57,584.30 |
| # orders B4702       | 5           | 5           | 4           | 4           |
| Permanent exp. X     | 0           | 0           | 100         | 100         |
| Permanent exp. Y (%) | 6.92%       | 6.92%       | 20.18%      | 20.18%      |

Table 17: Outsourcing: lead-time sensitivity analysis for B4702 under forecast demand

| Cost type         | Part 5b            | Part 6b (outsourcing, LT=1) |
|-------------------|--------------------|-----------------------------|
| Setup cost        | €132,000.00        | €130,500.00                 |
| Holding cost      | €360,656.60        | €280,147.80                 |
| Order cost B4702  | –                  | €2,500.00                   |
| Investment X      | €2,000.00          | €0.00                       |
| Investment Y      | €21,180.00         | €10,380.00                  |
| Overtime X        | €600.00            | €364.00                     |
| Overtime Y        | €0.00              | €3,432.00                   |
| Backorder cost    | €37,250.00         | €37,250.00                  |
| <b>Total cost</b> | <b>€553,686.60</b> | <b>€464,573.80</b>          |
| Service level     | 0.933              | 0.933                       |
| Fill rate         | 0.9752             | 0.9752                      |

Table 18: Outsourcing: cost comparison between Assignment 5b and 6b under realized demand

|                              | 5b (no outsourcing) | Outsourcing LT=1 | Outsourcing LT=2 | Outsourcing LT=3 | Out |
|------------------------------|---------------------|------------------|------------------|------------------|-----|
| Total cost (realized demand) | €553,686.60         | €464,573.80      | €486,013.80      | €497,438.10      |     |
| Difference with 5b           | –                   | €89,112.80       | €67,672.80       | €56,248.50       |     |
| Max. price premium per unit  | –                   | €1.13            | €0.86            | €0.71            |     |

Table 19: Outsourcing: break-even analysis of the maximum price premium per unit of B4702 under outsourcing, based on 78,847 units

## Appendix C: Extrapolation methods for extended horizon of 35 weeks

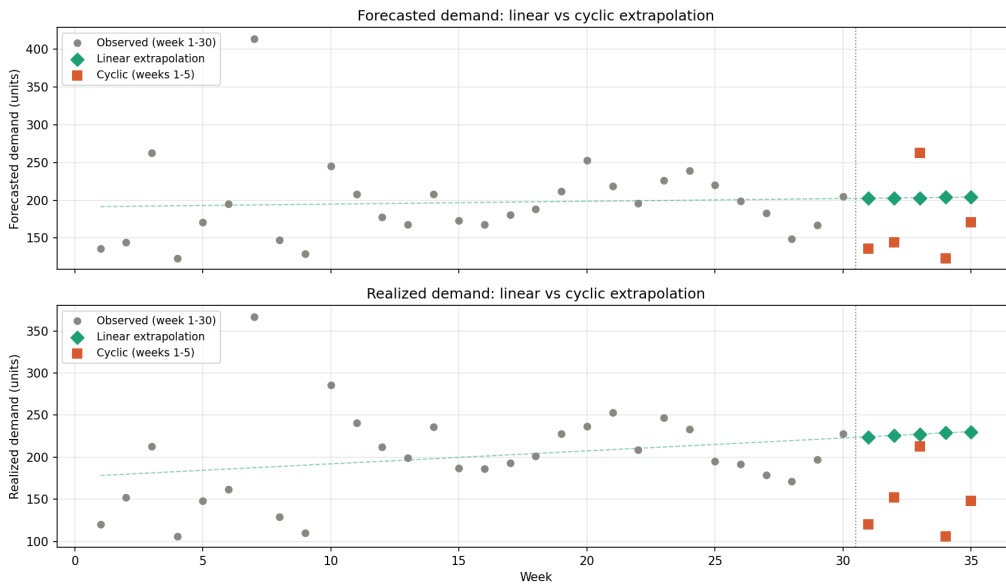


Figure 1: Comparison of extrapolation methods for extended planning horizon

## Appendix D: Sensitivity tables for capacity expansion

| Design  | Max cap X (units) | Forecast cost (€) | Realized cost (€) | #backorders | Service level |
|---------|-------------------|-------------------|-------------------|-------------|---------------|
| 1       | 100               | 512,123.10        | 555,733.10        | 149         | 93.3%         |
| 2       | 150               | 511,953.40        | 555,745.10        | 149         | 93.3%         |
| current | 200               | 510,108.10        | 553,718.10        | 149         | 93.3%         |
| 3       | 250               | 510,443.10        | 554,053.10        | 149         | 93.3%         |
| 4       | 300               | 510,843.10        | 554,453.10        | 149         | 93.3%         |

Table 20: Sensitivity table: forecast cost vs realized cost per capacity increase for workstation X

| Design  | Max cap Y (%) | Forecast cost (€) | Realized cost (€) | #backorders | Service level |
|---------|---------------|-------------------|-------------------|-------------|---------------|
| 1       | 4.1%          | 514,991.30        | 558,601.30        | 149         | 93.3%         |
| 2       | 9.1%          | 512,176.50        | 555,786.50        | 149         | 93.3%         |
| current | 14.1%         | 510,089.40        | 553,699.40        | 149         | 93.3%         |
| 3       | 19.1%         | 511,634.10        | 555,244.10        | 149         | 93.3%         |
| 4       | 24.1%         | 514,572.30        | 558,182.30        | 149         | 93.3%         |

Table 21: Sensitivity table: forecast cost vs realized cost per capacity increase for workstation Y